

Computational Economics

Lab: Introduction to Python

Python, development tools, environments, and notebooks

Yaolang Zhong yaolang.zhong@e.u-tokyo.ac.jp
Graduate School of Economics, The University of Tokyo

What Is Python?

Python is a programming language: a way to write instructions that a computer can carry out. We use it to work with data, perform calculations, and automate tasks.

- ▶ You write instructions as **code**, saved in files that you can edit and share.
- ▶ The **Python interpreter** is the software that executes those instructions. Installing Python gives you an interpreter.
- ▶ A **script** is a text file containing Python code, usually with the extension `.py`. Running the script asks the interpreter to execute it.

Writing code and running code are separate steps: editing a file does not execute it.

Where Do You Write Code?

An **integrated development environment (IDE)** brings together tools for writing, running, and inspecting code in one workspace.

- ▶ Open a project folder, browse its files, edit code, and see the output when you run it.
- ▶ We use **VS Code**. Its Python and Jupyter extensions add support for Python scripts and notebooks.
- ▶ Other options include **PyCharm** and **Cursor**.
- ▶ Select a **Python interpreter** for the project. The editor is your workspace; the selected interpreter runs your code.

Python Versions and Environments

- ▶ A **Python version** identifies a release of the language and interpreter. Projects may need a particular release to work with their code and tools.
- ▶ A project's **environment** is its Python setup: the interpreter it uses and the packages installed for it.
- ▶ A **virtual environment** gives a project its own set of installed packages. Changing one project's setup can then leave other projects' setups alone.
- ▶ When creating an environment, choose the Python interpreter it will use. In VS Code, select that environment when running the project.

To reproduce someone else's work, check both their Python version and their package versions.

What Is a Package?

A **package** provides reusable code written by others, so you can use existing tools without writing everything yourself.

- ▶ Common tools include **NumPy** for numerical arrays, **pandas** for data tables, and **Matplotlib** for plots.
- ▶ **Install** a package into the environment where you will run your code. **pip** is a tool for installing Python packages.
- ▶ **Import** makes a package's tools available to your code. Installing and importing are separate steps.
- ▶ Packages have versions too. A project's **dependencies** are the packages it needs; a file such as `requirements.txt` can list them and their versions.

What Is a Notebook?

A **Jupyter notebook** is a document that combines explanations, executable code, and results. It is saved as an `.ipynb` file.

- ▶ **Text cells** contain instructions, notes, and equations. **Code cells** contain Python instructions that you can run one cell at a time.
- ▶ A **kernel** is the running program that executes the cells. For Python, it uses a Python interpreter and its environment.
- ▶ The kernel remembers values between cell runs. Run cells from top to bottom so later cells receive the inputs they need.
- ▶ You can open a notebook in VS Code or use an online notebook service such as Google Colab.

Exercise: Open the Notebook in Colab

Google Colab lets you edit notebooks in your browser and run Python on a computer provided by Google. You can start this exercise without installing Python locally.

Open the course notebook in Colab

1. Sign in with your Google account and save your own copy in Drive.
2. Connect to a **runtime**: the computer session that runs your notebook. A standard CPU runtime is sufficient.
3. Read the notebook instructions and run its cells in order. The first code cell displays the Python version being used.
4. Complete the exercises: change an input, rerun the relevant cells, and record your observations.
5. Restart the runtime and run all cells. Save your completed notebook and download the `.ipynb` file to keep with your project.

References and Further Reading

1. **Feng (2026), AI for Economic Research: Dynamic Models, Language, and Agents. [Online course]**
 - ▶ Session 1.5: The Workbench: Python, VS Code, Git and GitHub
2. **Microsoft, Visual Studio Code Documentation. [Webpages]**
 - ▶ Getting Started with Python in VS Code
 - ▶ Jupyter Notebooks in VS Code
3. **Python Software Foundation, Python.org and Python Documentation. [Webpages]**
 - ▶ Download Python
 - ▶ Virtual Environments and Packages